PROJECT PROPOSAL

BLOCK 1D-2023-2024

Accident Hotspot Identification Dashboard

Names of the Students

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I. Introduction

The purpose of the project is to identify and visualize hotspots of various driving incidents in Breda, such as speeding, abrupt braking, sharp turns, and rapid acceleration. Using clustering algorithms and a machine learning model, the project aims to analyze historical incident data, label clusters based on predominant incident types, and predict these labels on unseen data. The ultimate objective is to develop a dashboard that displays these hotspots, enabling police and city planners to proactively allocate resources and implement safety measures.

The chosen topic is highly relevant and interesting because road safety is a critical public concern. Identifying and addressing hotspots of unsafe driving behaviors can significantly reduce accidents, save lives, and improve the overall quality of life in Breda. This project not only provides a data-driven approach to enhancing road safety but also offers a practical application of advanced machine learning techniques in a real-world context.

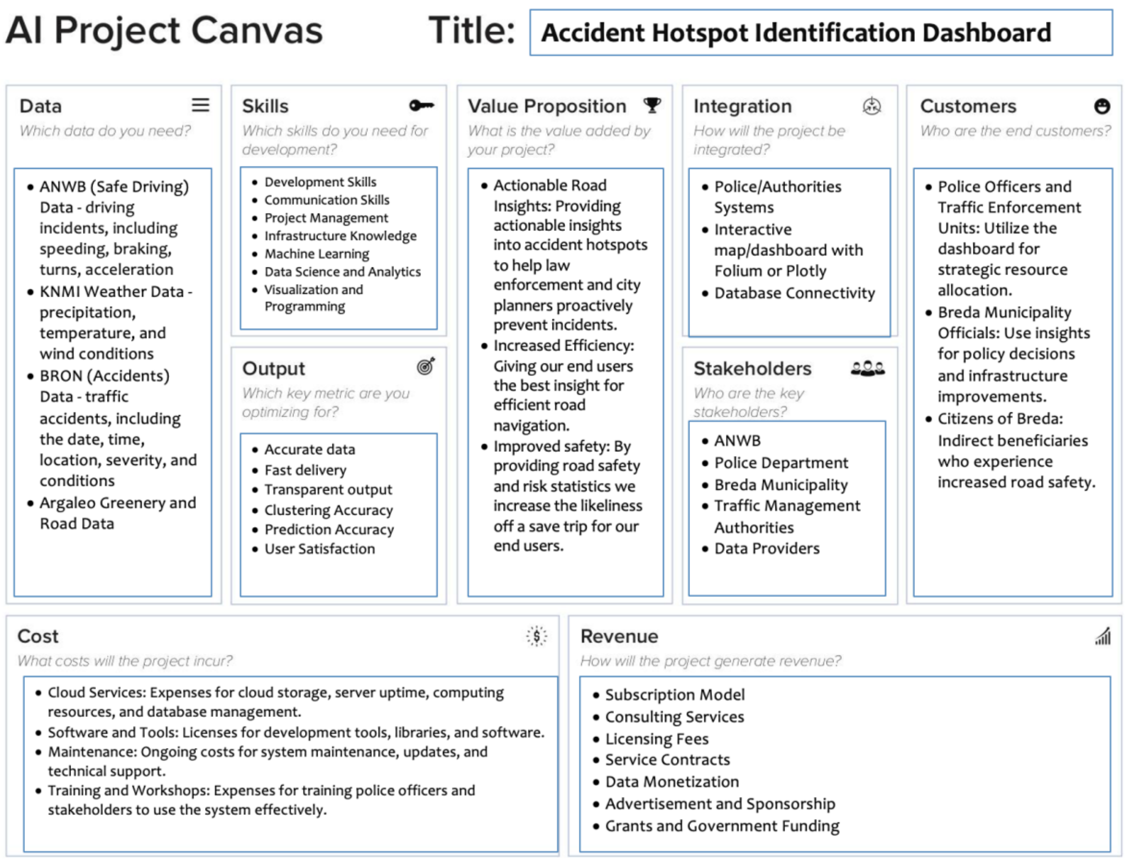
Applying machine learning techniques to this problem is significant because it allows for the extraction of meaningful patterns and insights from complex and diverse datasets. Clustering algorithms can effectively identify areas with high incidences of specific driving behaviors, while predictive models can forecast future hotspots based on new data. This data-driven approach enables more accurate and timely interventions, ultimately leading to a safer driving environment in Breda.

II. Problem Statement

Unsafe driving behaviors significantly contribute to road accidents, leading to injuries, fatalities, and property damage. Breda, like many urban areas, experiences various driving incidents that compromise road safety. The ANWB Safe Driving App provides data on these incidents, but there is no current system to analyze and predict where these incidents are likely to occur in the future. By leveraging historical incident data along with weather and traffic information, we can identify patterns and predict hotspots for these unsafe behaviors.

Therefore, our goal is to solve the problem of identifying and predicting hotspots of various unsafe driving behaviors in Breda. These behaviors include speeding, abrupt braking, sharp turns, and rapid acceleration. By using machine learning techniques, we will analyze historical incident data to cluster and label areas with high incidences of these behaviors and predict future hotspots based on unseen data.

Solving this problem is crucial for improving road safety and reducing accident rates in Breda. By accurately identifying and predicting hotspots of unsafe driving behaviors, police and city planners can strategically allocate resources and implement targeted interventions, such as speed cameras and traffic calming measures. This proactive approach not only enhances safety for drivers and pedestrians but also optimizes the use of public safety resources.



III. Data Description

For this project, we plan to use all the available datasets from the provided data lake, including ANWB (Driver Characteristics), KNMI (Weather), BRON (Accidents), and Argaleo (Greenery/Road). Each of these datasets offers a wealth of information critical for our analysis, particularly for clustering to understand the patterns and factors contributing to road safety incidents.

The ANWB dataset provides comprehensive details about driving incidents, such as the event's start and end times, location coordinates, speed measurements, and the type and severity of the incident. Key features for clustering from this dataset include "duration_seconds", "latitude", "longitude", "speed_kmh", "end_speed_kmh", "maxwaarde", "category", and "incident_severity". These attributes help in understanding the spatial and temporal distribution of incidents, as well as the driving behaviors associated with different types of incidents.

From the KNMI datasets, we will use weather-related variables, including precipitation, temperature, and wind data. Important features include "precipitation intensity", "air temperature", and "wind speed". These attributes are crucial for correlating weather conditions with driving incidents, helping us to determine if certain weather patterns contribute to higher risks of accidents.

The BRON dataset on accidents contains detailed information about each accident, such as the date and time, location, number of injuries or fatalities, and road conditions. Key features here include "date", "time", "latitude", "longitude", "number_of_injuries", and "road_conditions". This data allows us to analyze accident hotspots and the severity of accidents under varying conditions.

Lastly, the Argaleo datasets provide information on road characteristics and greenery, with features such as "road segment id", "road number", "road name", and "relative height". These attributes are essential for understanding how the physical and environmental characteristics of roads influence accident occurrences.

IV. Methodology

To tackle the problem of identifying and predicting accident hotspots in Breda, our approach involves a comprehensive machine learning workflow. This will include clustering and predictive modeling to analyze and forecast accident-prone areas based on various data attributes. We aim to combine historical accident data, weather information, and road characteristics to create a robust predictive system. The ultimate goal is to enhance road safety and assist authorities in proactive resource allocation.

The first step in our machine learning process is data collection. As mentioned in the previous section, we will use the data available in the data lake, including ANWB for driver characteristics, KNMI for weather data, BRON for historical accident records, and Argaleo for information on greenery and road conditions. These datasets provide a rich set of features necessary for our analysis.

Next, we will preprocess the data to handle missing values, remove duplicates, standardize date and time formats, and merge the different datasets. Data integration will involve merging datasets based on common keys, such as location coordinates and timestamps. Feature engineering will be crucial at this stage to create meaningful variables that capture the essence of each dataset. For instance, we might derive new features like average speed during different weather conditions or the severity of accidents based on road type.

After preprocessing, we will conduct exploratory data analysis (EDA) to understand data distributions, identify correlations, and visualize patterns. This step helps in gaining insights into the data and forming hypotheses about potential accident hotspots.

In the model selection phase, we will choose appropriate clustering algorithms for identifying accident hotspots and machine learning models for predicting future incidents. K-means clustering will be employed to identify spatial and temporal clusters of accidents, as it is effective in finding distinct groups based on similarity. For prediction, we will use either Random Forest or Gradient Boosting Machine (GBM) due to their robustness in handling complex, non-linear relationships and providing high accuracy.

Training the clustering model involves using historical data to identify patterns and clusters of accident occurrences. The predictive model will then be trained using features derived from these clusters along with other significant attributes such as weather conditions and road characteristics. Model evaluation will be conducted using metrics like the silhouette score for clustering and accuracy, precision, recall, and F1-score for the predictive model.

Once the models are trained and evaluated, we will deploy them into a dashboard that visualizes accident hotspots and provides predictions on new unseen data. This dashboard will enable authorities to monitor accident-prone areas and allocate resources more effectively.

The chosen clustering model, K-means, is suitable for this task as it identifies accident hotspots based on spatial and temporal data such as latitude, longitude, time of day, and weather conditions. The predictive model, either Random Forest or GBM, is selected for its ability to handle complex interactions between variables and provide reliable predictions of future accidents.

The architecture of the clustering model includes input features like latitude, longitude, time of day, incident severity, and weather conditions, with the output being a set of clusters representing accident hotspots. The predictive model takes these cluster labels along with historical accident frequencies, weather data, and road characteristics as input features to predict the likelihood of accidents in specific areas.

There are several risks associated with this AI system, including data quality issues, model accuracy concerns, and potential bias. Poor data quality could lead to inaccurate predictions, while model inaccuracies could result in inefficient resource allocation and potential safety risks. Additionally, there is a risk of the models inadvertently learning and propagating biases present in the historical data. To mitigate these risks, we will implement rigorous data validation, model evaluation, and bias detection techniques, along with regular model retraining with up-to-date data.

Lastly, we must address legal obligations such as data privacy and transparency. Compliance with GDPR is essential, requiring us to anonymize personal data and obtain necessary consents for data usage. Transparency in data collection, processing, and model decision-making processes is crucial to maintain trust. Ethical use of the system is paramount, particularly in avoiding discriminatory practices and ensuring public safety. We will conduct regular audits of our data handling practices, implement robust data security measures, and maintain open communication with stakeholders to address any legal or ethical concerns promptly.

V. Project Timeline

The project will be structured into specific tasks with designated timeframes to ensure a systematic and efficient approach to achieving our objectives. We will utilize various tools and platforms such as Teams meetings, DataLab sessions, and Trello for task organization and coordination.

Project Planning and Setup will be completed by the end of this week (week 3). During this phase, we will define the project scope and objectives, set up project management tools, and assign roles and responsibilities to each team member. This foundational work is crucial to ensure that the project is well-organized and that everyone is aligned with the project goals.

Data Collection and Integration, Data Pre-processing, and Exploratory Data Analysis (EDA) will be conducted by the end of week 4. In this phase, we will gather data from various sources including ANWB, KNMI, BRON, and Argaleo. After collecting the data, we will integrate and merge these datasets, followed by cleaning and preprocessing to handle any missing values, duplicates, and normalization. The EDA will involve analyzing data distributions, identifying correlations, and visualizing data patterns to gain insights and guide subsequent model development.

Model Selection and Development, Model Training and Evaluation, and Dashboard Development will be completed by the end of week 5. We will select appropriate clustering algorithms, such as K-means, to identify accident hotspots and predictive models, like Random Forest or Gradient Boosting Machine, to forecast future accidents. These models will be trained on historical data, and their performance will be evaluated using metrics like accuracy, precision, recall, and F1-score. Simultaneously, we will design and develop a dashboard that integrates these models and provides real-time data visualization.

Model Deployment and Testing and Validation will take place during week 6. In this phase, the clustering and predictive models will be deployed, and the dashboard will be made accessible to end-users. We will conduct rigorous testing to ensure functionality and accuracy, validating the models and dashboard with real-time data to confirm their effectiveness.

Throughout the entire project, we will engage in Documentation and Reporting, Project Review and Optimization, and Team Meetings and Coordination. These activities will be ongoing and will involve documenting our processes, methodologies, and findings, preparing final reports and presentations, and regularly reviewing project outcomes with stakeholders. We will use feedback to identify areas for improvement and optimization, implementing necessary adjustments. Regular team meetings using Teams and DataLab sessions, along with task tracking on Trello, will ensure continuous communication and coordination among team members.

By following this structured timeline, we aim to achieve a high level of organization and efficiency, ensuring the successful completion of the project.

VI. Reference

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